AHMET FURKAN YORULMAZ

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Profile

Fourth-year Software Engineering student at Istanbul Atlas University whose interest in technology started in high school through countless hours spent with computers. Passionate about building innovative products and software that make a real difference in people's lives. Has delivered projects across a wide range of domains — from AI-integrated systems and mobile applications to embedded hardware — including a graduation thesis on autonomous vehicle systems, which reflects a long-standing passion for automotive technology. Outside of code, likes experimenting with microcontrollers as a hobby.

Education

ISTANBUL ATLAS UNIVERSITY – Software Engineering (English) — **September 2022 – Present**

REPUBLIC OF TÜRKİYE MINISTRY OF INDUSTRY AND TECHNOLOGY & T3 FOUNDATION
– Autonomous Driving Technologies Program — **January 2026 – March 2026**

GOOGLE & ARTIFICIAL INTELLIGENCE AND TECHNOLOGY ACADEMY
– Artificial Intelligence Scholar — **January 2026 – Present**

Work Experience

DATIVA – AI Engineering Intern — **October 2025 – May 2026**

Contributed to the development of AI-driven solutions for corporate clients by optimizing large-scale document analysis workflows using the Gemini File Search Tool and indexing techniques. Built automated pipelines that processed hundreds of pages of tender and technical documents, generating standardized JSON-based analyses aligned with company evaluation criteria.

- Built an automated analysis pipeline that indexed and scanned hundreds of pages of tender documents within a single global store using the Gemini File Search Tool.
- Designed a flexible background-processing system that converts Word documents into PDF using docx2pdf and tempfile, enabling different file formats to be handled through a unified workflow.
- Created detailed and rule-based system instructions and JSON schemas to extract technical requirements from multilingual tender documents and map them into predefined structured fields.

Additionally, developed an AI-powered speech-to-text and video question-answering system using FFmpeg, Faster-Whisper, and the OpenAI GPT-4 Mini API. The system automatically detected video language, produced timestamped JSON and SRT outputs, and enabled users to ask natural-language questions about industrial training videos through a hybrid LLM-, keyword-, and semantic-based search approach, returning precise answers with exact video timestamps.

- Implemented a Python-based speech-to-text pipeline that detects spoken language and generates time-aligned JSON and SRT transcripts from video content.
- Applied a hybrid retrieval architecture combining LLM reasoning, keyword-based search, and semantic similarity to enable accurate question answering over video transcripts.
- Integrated the OpenAI GPT-4 Mini API to generate context-aware answers and return exact video, minute, and second references.
- Optimized the system for industrial machine training videos, enabling fast and precise access to technical information.

TNC GROUP – Project Management Intern — **February 2026 – March 2026**

Completed a structured 5-week project management internship program at TNC Group, combining weekly training sessions with hands-on project deliverables across four core disciplines.

- Studied fundamental and applied project management methodologies, learning how to plan, execute, and monitor projects within organizational constraints.
- Gained an understanding of IT Law and legal frameworks governing digital systems, data privacy, and software-related compliance requirements.
- Explored management and organizational theory, including team structures, decision-making processes, and coordination strategies in corporate environments.
- Completed a module on Artificial Intelligence and Data Science, examining AI applications in business contexts and data-driven decision-making approaches.
- Delivered a separate project each week aligned with the training topic, applying theoretical knowledge to practical scenarios throughout the program.

TÜPRAŞ – Software Development Intern **July 2025 – August 2025**

Rotated across the Cybersecurity, Software Development, and Operational Systems departments, delivering hands-on projects in network security, AI-powered tooling, and mobile application development.

- [Cybersecurity Dept.] Built Scanpro, a network security scanner in Flask and Python — established a cross-machine SSH connection via PuTTY, then extracted TCP/UDP port data and service banner information from a remote virtual machine running on a separate host.
- [Cybersecurity Dept.] Implemented automated subdomain enumeration to discover all live subdomains of target websites, extending the scanner to run a full security analysis across every discovered subdomain.
- [Software Development Dept.] Developed an internal AI chatbot prototype using Next.js and React — designed to assist management with task distribution and team coordination across departments.
- [Software Development Dept.] Collaborated with a mentor to design and prototype Tripzy, a React Native mobile application with AI-powered travel route generation, to strengthen mobile development capabilities.
- [Operational Systems Dept.] Gained hands-on exposure to field operations and site management workflows, observing how large-scale industrial operations are planned and coordinated on the ground.

TZİ CONSULTING – Web Developer (Full Time) **October 2024 – July 2025**

- Designed a website for the company with WordPress and developed modern interfaces to improve user experience.
- Enabled multilingual support with TranslatePress, making the site easily translatable into different languages.
- Implemented POS integration, enabling fast and secure payment transactions.
- Created visually rich, responsive and dynamic page designs with Elementor.

ETERNA VISIO – Software Engineer Intern **July 2024 – August 2024**

- Developed an AI chatbot for a web platform using Next.js 13, React, Tailwind, Prisma and Stripe.
- Optimized AI performance using Next.js 13's SSR and SSG features.
- Built a user-friendly and dynamic interface using a component-based structure in React.
- Designed a modern, flexible and fast user interface for the AI platform using Tailwind CSS.
- Optimized database management with Prisma, accelerating data storage and querying processes of the AI model.
- Enabled users to pay for AI services safely and seamlessly through Stripe integration.
- Integrated the OpenAI API to provide smart responses and automatic content generation features.

Projects

Autonomous Lane-Following Robot Vehicle | Graduation Thesis

- Engineered a fully autonomous 4-wheeled robot vehicle capable of real-time lane following, corner navigation, and traffic light response on a 250×250 cm closed-loop track.
- Designed a dual-processor architecture with Raspberry Pi 4 handling image processing and decision-making, and Arduino Uno R3 managing real-time motor control via GPIO bit-bang UART communication.
- Implemented HSV color space filtering and connected component (blob) analysis with OpenCV to distinguish dashed inner and solid outer white lane markings, keeping the vehicle centered in its lane.
- Built corner detection using horizontal run-length analysis — vehicle decelerates, executes a tank turn, and visually confirms lane alignment before resuming.

- Architected a 7-state Finite State Machine (FSM) governing all driving logic, with Exponential Moving Average (EMA) for adaptive lane-width learning and a live MJPEG stream (port 8080) for real-time monitoring.
- **Technologies used:** Python 3, OpenCV, Raspberry Pi 4, Arduino Uno R3, L298N Motor Driver, Raspberry Pi Camera Module, pigpio, Picamera2, NumPy

Tripzy – AI Integrated Tourist Assistant Mobile Application (Ongoing Project)

- A full-stack mobile application that automatically generates personalized holiday routes based on destination, travel duration, group size, and budget, with AI-powered Q&A and a community travel-sharing section.
- Developed a RESTful backend using Spring Boot 3 and Java 17, implementing JWT-based authentication (register/login endpoints) secured with Spring Security.
- Designed a MySQL database schema with Spring Data JPA, managing user entities with validation, timestamps, and account status tracking.
- Documented all API endpoints using OpenAPI/Swagger for clean integration with the mobile frontend.
- Built the React Native mobile frontend with OpenAI-powered route generation, public transportation suggestions, and an "Explore" section for community travel plan sharing.
- **Technologies used:** React Native, Java, Spring Boot 3, Spring Security, JWT, MySQL, Spring Data JPA, OpenAPI/Swagger, Maven, OpenAI API

KOOPILOT – AI-Powered E-Commerce Management Dashboard | AI Hackathon

- An AI-driven operations dashboard for small businesses and producer cooperatives, centralizing product stock, orders, customer messaging, and sales analytics in a single interface — with live production deployment on Railway.
- Developed a Flask 3.1 backend with a clean service-layer architecture, connecting to a Supabase PostgreSQL database via psycopg2, with dedicated modules for data queries, AI integration, and dashboard analytics.
- Integrated Google Gemini API to power a context-aware AI assistant that answers natural-language business queries using live data — products, stock levels, recent orders, and pending customer messages injected into each prompt.
- Built interactive sales reports (daily/weekly/monthly) with dynamic Chart.js visualizations and a full-featured customer DM system with complete conversation history.
- Deployed to production on Railway using Gunicorn as the WSGI server.
- **Technologies used:** Python 3, Flask 3.1, PostgreSQL, Supabase, psycopg2, Google Gemini API, Jinja2, Chart.js, Gunicorn, Vanilla JS

TaskFlow – Trello-Style Kanban Project Management Board

- Built a full-stack Kanban board application with drag-and-drop card and column management, user authentication, and activity logging — deployed on Vercel.
- Implemented fractional indexing for card and column ordering so that any drag-and-drop reorder requires a single database UPDATE, avoiding costly full-list renumbering.
- Architected optimistic UI updates with Zustand — drag events instantly update the local store and automatically roll back to a snapshot on API failure.
- Configured dnd-kit with PointerSensor (desktop), TouchSensor (mobile, 220 ms long-press), and KeyboardSensor (accessibility), with a DragOverlay portal ensuring smooth rendering during drag.
- Secured all REST endpoints with Auth.js v5 (Credentials + JWT sessions) and per-resource ownership checks — unauthorized board access returns 404.
- **Technologies used:** Next.js 15, React 19, TypeScript, PostgreSQL (Neon), Prisma ORM, Auth.js v5, dnd-kit, Zustand, Zod, Tailwind CSS, Vercel

AI Chatbot – Context-Aware AI Chat Web Application

- Developed a full-stack AI chatbot web application supporting persistent conversation history, user authentication, and multi-file uploads (up to 5 files per message).
- Built with Next.js 15 App Router and TypeScript, using Prisma ORM for PostgreSQL-backed chat history persistence and Auth.js middleware for session-protected routes.
- Designed a component-driven UI with shadcn/ui and Tailwind CSS, providing a clean and responsive chat interface with file attachment support.
- **Technologies used:** Next.js 15, TypeScript, PostgreSQL, Prisma ORM, Auth.js, shadcn/ui, Tailwind CSS

Scanpro – Web Based Security Scanner Application

- Built a web-based network security scanner using Flask and Python; deployed across two separate machines connected via PuTTY (SSH) to simulate a real-world remote scanning environment.
- Implemented TCP and UDP port scanning with service banner grabbing to identify running services and potential exposure points on target hosts.
- Developed automated subdomain enumeration that discovers all live subdomains of a given domain and runs a full security scan on each one, surfacing hidden attack surfaces.
- Presented scan results through a structured web interface built with React.js, displaying IP addresses, open ports, protocols, and detected vulnerabilities in a readable report format.
- **Technologies used:** Flask, Python, Nmap, React.js, Ubuntu 22.04, PuTTY, JSON, HTML/CSS

Smart Feeding Device

- Provides pets with food and water in desired amounts and at desired times via a digital screen and mobile application, with the possibility of veterinary control.
- **Technologies used:** ESP32, C/C++/Python, 3D Printer, Autodesk Fusion 360, AutoCAD

Skills

Languages: Python, Java, TypeScript, JavaScript, C, HTML, CSS

Frameworks & Libraries: Next.js, React.js, React Native, Spring Boot, Flask, OpenCV, Tailwind CSS

Databases: PostgreSQL, MySQL, SQL, Prisma ORM

AI & APIs: OpenAI API, Google Gemini API

Tools & DevOps: Git, Docker, Nmap, PuTTY, WordPress, Autodesk Fusion 360

Hardware & Embedded: Arduino, Raspberry Pi, ESP32, Sensors, Soldering

Certificates

Issuing Organization: Multi Group / Multi Academy
Certificate Name: Zero2End Machine Learning Certificate of Participation
Date: 02 January 2026

Issuing Organization: Coursera / Google
Certificate Name: Foundations of Project Management
Date: 28 January 2026

Issuing Organization: Coursera / Google
Certificate Name: Project Initiation: Starting a Successful Project
Date: 27 February 2026

Issuing Organization: Borusan / Toptalent Business School
Certificate Name: Borusan Technology School Completion Certificate
Date: 01 March 2026

Languages

English (Fluent speaking and writing), Turkish (Native)